

Spectral Covariance Cleaning, Portfolio Breadth, and Market Proxies: Evidence from Mexican Equities

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Abstract

Does nonlinear RMT covariance cleaning reduce out-of-sample minimum-variance portfolio risk without diminishing allocation breadth in Mexican equities, and how does the observed relationship vary across market proxies and asset-to-observation ratios? We compare sample, eigenvalue-clipped, Ledoit–Wolf, and rotationally invariant (RIE) covariance estimators on a common, chronological holdout from a 38-stock Bolsa Mexicana de Valores panel. On 529 test days, daily realized volatility is 0.6794% per day for RIE and 0.6884% for the sample estimator, a 1.31% relative difference. This volatility comparison is descriptive: the paired 20-day moving-block bootstrap for the annualized Sharpe difference is +0.011 (95% interval $[-0.021, 0.057]$; one-sided $p = 0.186$). RIE’s inverse-Herfindahl effective holdings count is 9.86 versus 9.36, while Shannon effective breadth is 26.92 for both; allocation-breadth conclusions depend on the measure. Two provisional VOM-labelled Bloomberg proxies show higher conditional hold-out volatility above training-period 90th-percentile thresholds, but their economic definitions are undocumented and the comparisons are descriptive. In a recent 25-stock, 100-return panel ($q = 0.25$), eigenvalue clipping reduces the correlation condition number from 24.34 to 5.33. In ten deliberately short $q = 1.9$ windows, this RIE implementation is unstable, with 144.43% mean annualized realized volatility. The evidence supports a conditional, limited case for spectral regularization, not universal estimator dominance or a market-wide claim about investor diversity.

Keywords: random matrix theory; covariance estimation; Mexican equity market; minimum-variance portfolio; covariance conditioning; portfolio concentration; out-of-sample evaluation.

1 Introduction

The sample covariance matrix is a noisy input to portfolio optimization when the number of assets is not negligible relative to the return-history length. Because minimum-variance weights depend on its inverse, errors in the smallest sample eigenvalues can be magnified in the resulting positions. Random matrix theory (RMT) provides principled spectral corrections, including nonlinear shrinkage estimators that preserve the sample eigenvectors while modifying eigenvalues [1, 7, 2, 3, 6]. Early empirical work documented noise in financial correlation spectra, while subsequent studies connected cleaning to portfolio optimization [4, 5]. Whether these corrections remain useful across market conditions while maintaining broad allocations is an empirical question for emerging equity markets.

Research question. *Does nonlinear RMT spectral covariance cleaning lower out-of-sample minimum-variance portfolio risk without reducing allocation breadth in Mexican equities, and how do the observed risk and breadth outcomes vary with market proxies and the asset-to-observation ratio $q = N/T$?*

The article answers this question with one common chronological holdout for the primary estimator comparison, two separately exported Bloomberg market-indicator series for descriptive conditioning, and a recent high-dimensional cross-section. Each supplementary dataset has a distinct role; none is pooled into the primary panel.

The contribution is deliberately empirical and bounded: (i) a common-date comparison separates small realized-volatility differences from Sharpe-ratio uncertainty; (ii) two distinct allocation-breadth measures reveal that a diversity conclusion is metric-dependent; and (iii) high-dimensional checks document improved conditioning at $q = 0.25$ but instability of this particular unmodified RIE implementation at $q = 1.9$.

2 Data and methods

2.1 Primary BMV panel

The primary panel comprises 38 Mexican equities and 2,644 daily log-return observations, from 18 July 2011 through 29 September 2025, derived from adjusted Economática prices. The tickers are MFRISCOA-1, RA, SIMECB, FUNO11, ICHB, GENTERA, SORIANAB, ELEKTRA, CHDRAUIB, LIVEPOLC-1, AUTLANB, HERDEZ, GAPB, OMAB, MEGACPO, PINFRA, LABB, ALSEA, ARA, ASURB, AMXB, AC, AXTELCPO, KIMBERA, CEMEXCPO, BIMBOA, BOLSA, GCARSOA1, FEMSAUBD, GFNORTEO, GFINBURO, ORBIA, GRUMAB, GMEXICOB, PE&OLES, TLEVISACPO, SIGMAFA, and WALMEX. We reserve the final 529 observations (3 May 2021–29 September 2025) as a chronological holdout and estimate the covariance matrix on the preceding 2,115 observations. Thus $q = N/T_{\text{train}} = 38/2115 \approx 0.0180$. This common split is used for the estimator comparisons in Sections 3.1–3.4; it avoids mixing results computed on different test windows.

Daily log returns are computed from adjusted prices. We compare the sample covariance (SM), Marčenko–Pastur eigenvalue clipping (CL), Ledoit–Wolf linear shrinkage (LW), and analytic nonlinear rotationally-invariant shrinkage (RIE). CL standardizes to a correlation matrix, retains sample eigenvalues above the Marčenko–Pastur upper edge, replaces the remaining bulk eigenvalues by their average, and restores the original marginal scales. For sample aspect ratio $q = N/T < 1$, the theoretical noise support is $[\lambda_-, \lambda_+]$ with $\lambda_{\pm} = (1 \pm \sqrt{q})^2$ under the Marčenko–Pastur benchmark. LW is the standard identity-target linear shrinkage estimator. RIE applies kernel-based analytic nonlinear shrinkage to the sample eigenvalues while retaining the sample eigenvectors. The RIE implementation is designed for $q < 1$; this article does not claim that the implementation is valid for $q \geq 1$.

For each estimate $\hat{\Sigma}$, the unconstrained global minimum-variance weights are

$$\hat{w} = \frac{\hat{\Sigma}^{-1}\mathbf{1}}{\mathbf{1}^\top \hat{\Sigma}^{-1}\mathbf{1}}, \quad \mathbf{1}^\top \hat{w} = 1.$$

All four methods use the same training dates, test dates, return units, and fully invested constraint. The covariance estimates and weights are fit once on the training sample and held fixed through the full chronological test window; this is a static holdout comparison, not a rolling rebalancing strategy. We do not include transaction costs, turnover, liquidity limits, market impact, short-sale constraints, or leverage constraints.

The primary outcomes are annualized Sharpe ratio (using an assumed 8% annual risk-free rate), realized daily volatility on the holdout, and two allocation-breadth measures. The assumed risk-free rate is fixed across all estimators; results under alternative risk-free rates are not reported. The inverse-Herfindahl effective holdings count is $NEA = 1/\sum_i w_i^2$. Shannon breadth is $\exp(H)$, where $H = -\sum_i p_i \log p_i$ and $p_i = |w_i|/\sum_j |w_j|$; absolute weights are used because the unconstrained portfolios may contain short positions. With short positions, neither metric should be read literally as a count of positive holdings: both are portfolio-level summaries and say nothing about the diversity of market participants.

For a portfolio return $r_{p,t}$, the annualized Sharpe is computed as $\sqrt{252}(\bar{r}_p - r_f/252)/s_p$, with daily sample standard deviation s_p . The risk comparison is reported in percent per trading day. The primary Sharpe-difference interval is estimated by a paired moving-block bootstrap with 5,000 replications and 20-day blocks, using the same resampled date indices for both portfolios. The reported one-sided p -value tests whether the Sharpe difference is non-positive; no corresponding confidence interval for the volatility difference is available. Accordingly, the volatility gap is descriptive and is not presented as an inferentially established effect.

2.2 Supplementary market data and sample construction

Table 1 records the datasets used. The primary 38-stock panel supports the estimator comparison. A separate Bloomberg export supplies two VOM-labelled daily series, the NAFTRAC field PR093, and the USD/MXN exchange rate. A second Bloomberg workbook supplies daily fields for a recent 25-stock cross-section. These supplementary samples are used for descriptive regime conditioning and an aspect-ratio boundary check, respectively; they are not pooled with the primary return panel.

Table 1: Available data and role in the analysis.

Source	Observed material	Availability	Role here
BMV–Economática	38 equities; 2,644 daily returns, 2011–2025	Derived panel; raw licensed prices not redistributable	Primary common holdout and estimator comparison; universe selection may induce survivorship or selection bias
Bloomberg VOM/FX export	Two VOM-labelled series, NAFTRAC PR093, USD/MXN; 2000–2026	Daily CSV export; some dates absent	Convert VOM levels to MXN and condition holdout risks on training-period thresholds
Bloomberg BMV/BIVA workbook	275 instrument tabs and daily fields; 2000–2026	Licensed XLSX export	Recompute the matched $N = 25$, $T = 100$, $q = 0.25$ check using gross total returns
Fama–French 5 factors	Daily U.S. factors, matched to BMV dates	Public factor file	Secondary pricing-model check

Currency treatment and VOM-regime construction. The Bloomberg export identifies VOMXCUS\$ Index and VOMXGUS\$ Index as USD-denominated series and includes daily MXN BGN Curncy. Following the export note, we convert each observed VOM level to pesos by multiplying by the same-date USD/MXN rate. We use no cross-series comparison of raw levels: in particular, NAFTRAC’s exported field is PR093, not a closing price, so it is not treated as a return or compared numerically with either VOM series. For each VOM series separately, the high-state threshold is its 90th percentile in the primary training period (18 July 2011–30 April 2021), and this fixed

threshold is applied to matched holdout dates. There are 506 matched days (out of 529); the final VOM data date is 17 September 2025. The CSV does not include a Bloomberg field description for the VOM tickers; their interpretation as market-stress indicators is therefore provisional and should be checked against the terminal’s field metadata before external publication.

Recent high-dimensional cross-section. The recent Bloomberg workbook has an issuer list and 275 instrument-specific tabs with daily market fields. For an independent high-dimensional check, we match the 25 securities in the prespecified diagnostic one-to-one to workbook tabs: WALMEX, KIMBERA, GCARSOA1, GMEXICOB, BIMBOA, SIGMAFA, SORIANAB, LIVEPOLC, KUOA, KUOB, PE&OLES, GFNORTEO, VITROA, ARA, GRUMAB, GCC, ICHB, HERDEZ, ALSEA, ORBIA, SIMECB, CMRB, MEDICAB, VASCONI, and GFINBURO. We use Bloomberg’s `TOT_RETURN_INDEX_GROSS_DVDS` field rather than unadjusted prices. The common level window is 12 May–29 September 2026 (101 levels); log returns run from 13 May through 29 September (100 observations), giving $q = 25/100 = 0.25$. This window is used to re-estimate the sample correlation spectrum and conditioning; unlike the primary panel it provides no subsequent holdout. The fixed 25-ticker set is not a random sample of the 275-tab universe and is used only for this diagnostic.

2.3 Robustness and boundary checks

For a non-Gaussian sensitivity check, each training-series marginal is rank-transformed to normal scores, rescaled to its original volatility, and used to estimate the correlation structure; all portfolios are still evaluated on the untransformed holdout returns. This is a rank-based Gaussianization, not a fitted multivariate Student- t copula.

We also report (i) the risk of the four primary portfolios conditional on the two converted VOM-series thresholds; (ii) the re-estimated condition-number comparison at $q = 0.25$; (iii) a rolling real-data $q = 1.9$ check, where $T < N$; and (iv) regressions of the four primary holdout portfolio excess-return series on the public Fama–French five factors. These are separate validation exercises, not additional observations in the main estimator comparison.

3 Results

3.1 Common-holdout comparison

Table 2 reports results recalculated on the same 2,115/529 training/test split using the project’s shared estimator implementations. RIE has the lowest realized volatility, 0.679% per day versus 0.688% for SM, a relative difference of approximately 1.3%. The difference in Sharpe ratios is small and imprecise: RIE is -0.233 versus -0.244 for SM. The two intermediate estimators show why the result should not be described as a monotone ordering by estimator sophistication: LW has slightly lower realized volatility, and clipping has a materially less negative point-estimate Sharpe on this particular holdout.

Figure 1 visualizes the four holdout volatilities. Relative to SM, the RIE point estimate is lower by 0.0090 percentage points per trading day, or 1.31%; the absolute daily difference is small. The unadjusted daily Sharpe estimates are negative under the assumed 8% risk-free rate, and clipping has the least negative estimate. Estimator rankings therefore depend on the selected loss or performance statistic.

RIE’s NEA is about 5.3% higher than SM’s on this split. The Shannon breadth values, however, are almost identical (26.92 for both), and CL has a marginally lower NEA but a slightly lower

Table 2: Minimum-variance portfolios on the common BMV holdout. Volatility is percent per trading day; Sharpe is annualized.

Estimator	In-sample vol.	Holdout vol.	Sharpe	NEA	$\exp(H)$
Sample (SM)	0.7077	0.6884	−0.244	9.36	26.92
Clipping (CL)	0.7182	0.6860	−0.110	9.32	26.54
Ledoit–Wolf (LW)	0.7059	0.6837	−0.235	9.75	26.93
RIE	0.7103	0.6794	−0.233	9.86	26.92

All estimators are close on the common 529-day holdout

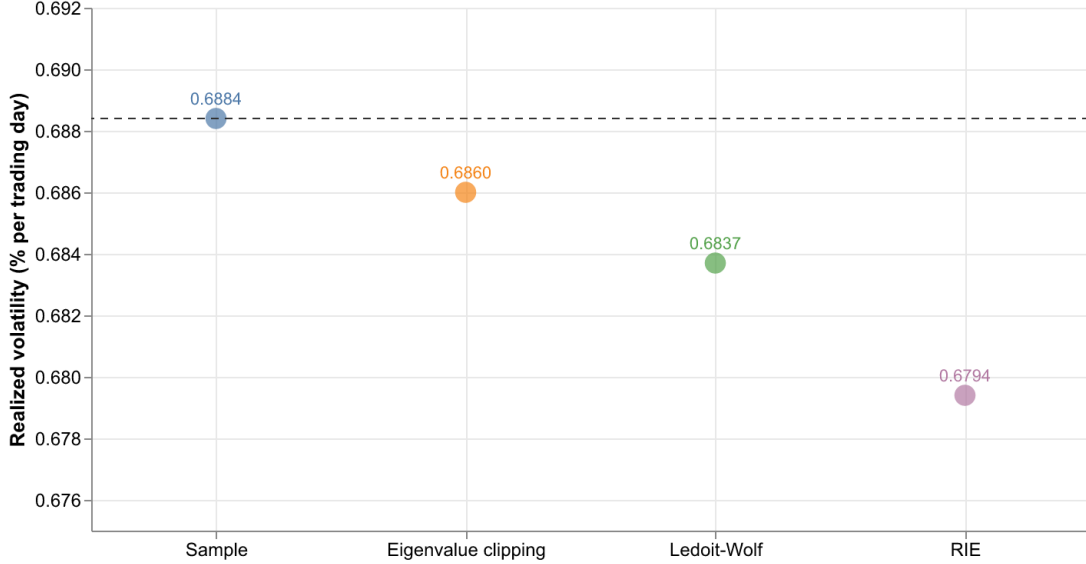


Figure 1: Realized daily volatility on the common 529-day holdout. The dashed line marks the sample-covariance estimate. The vertical axis is truncated to show the small differences; this is a descriptive comparison and no volatility-difference confidence interval is available.

Shannon breadth. The conclusion is therefore metric-dependent: spectral cleaning does not reduce either measure relative to SM here, but the evidence for a practically large increase in diversity is weak. This distinction is important when connecting portfolio concentration to the broader concept of decision heterogeneity.

Figure 2 puts both breadth statistics on separate scales. RIE’s higher NEA does not translate into a higher Shannon breadth, and eigenvalue clipping has a lower NEA than SM. The defensible conclusion is that RIE did not reduce either reported breadth measure relative to SM in this sample, not that it produces a broad or diversified portfolio in an absolute sense.

The paired moving-block bootstrap, with 5,000 resamples and 20-day blocks, gives an observed RIE–SM Sharpe difference of +0.011, a 95% interval $[-0.021, 0.057]$, and one-sided $p = 0.186$. We cannot reject equal Sharpe performance on this holdout. The volatility difference is descriptive and is not presented as a statistically significant treatment effect.

3.2 Distributional sensitivity and market-regime validation

In the rank-Gaussianized sensitivity, the ordering by holdout-to-training volatility ratio remains CL, RIE, LW, SM; RIE does not rank first. Table 3 gives the corresponding realized holdout

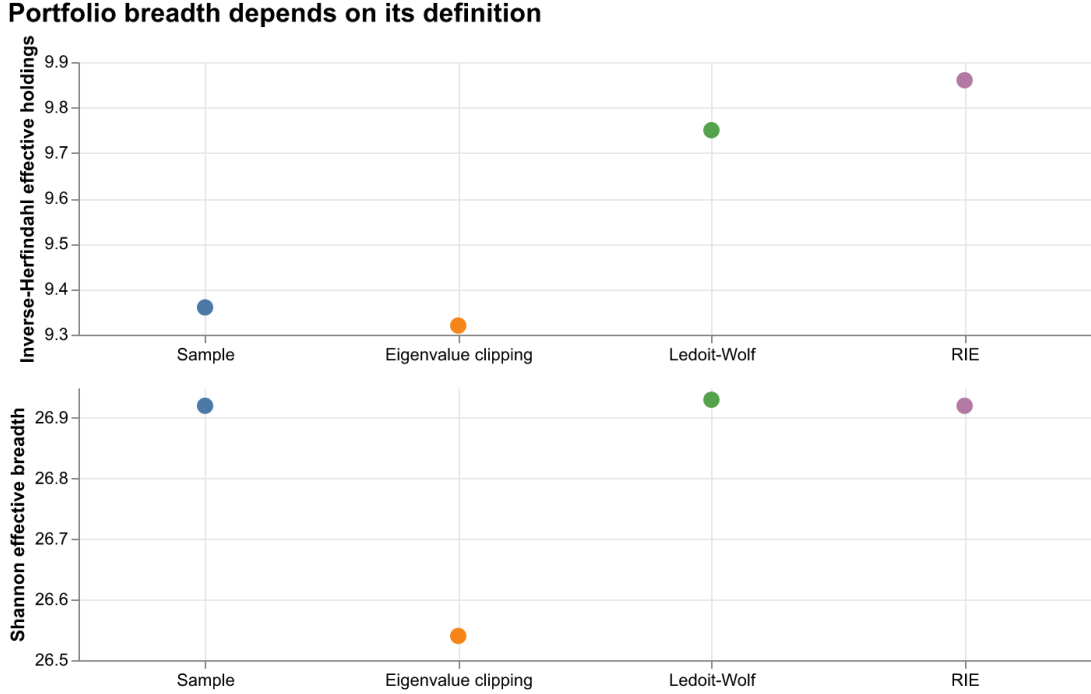


Figure 2: Two allocation-breadth summaries for the four optimized portfolios. Each panel has its own vertical scale. Because short sales are allowed, these statistics summarize weight concentration and are not literal counts of long positions.

volatilities.

Table 3: Holdout volatility under raw and rank-Gaussianized training marginals. Evaluation always uses the same untransformed holdout returns.

Estimator	Raw training returns	Rank-Gaussianized training returns
SM	0.6884	0.6942
CL	0.6860	0.6887
LW	0.6837	0.6919
RIE	0.6794	0.6870

For the primary training panel, per-asset excess kurtosis ranges from 1.18 to 20.20, with median 5.34, confirming substantial non-Gaussian tails. The rank-Gaussianized sensitivity in Table 3 preserves the lower holdout volatility of RIE relative to SM, but places CL first by the holdout-to-training risk ratio. This robustness result is descriptive, and does not establish that one estimator dominates under every distribution or loss criterion.

Table 4 compares the same static portfolios on matched holdout days above and below each converted VOM-series threshold. For **VOMXCUS\$**, 188 matched days exceed the fixed training-period 90th-percentile threshold and 318 fall below it; realized portfolio volatility is 30–32% higher in the high state for all four estimators. RIE’s standard deviation is marginally lowest in both states. For **VOMXGUS\$**, the high-state volatility increase is smaller and the estimator differences are likewise modest. These conditional standard deviations are descriptive rather than causal or inferential tests; the Bloomberg export lacks field documentation needed to assign a definitive economic interpretation to the VOM series. Figure 3 shows the full set of conditional standard deviations and

makes clear that the size of the high/low difference varies by proxy.

Table 4: Holdout daily portfolio volatility by converted VOM-series state (percent per day). Thresholds are fixed at each series’ training-period 90th percentile.

Proxy	State	Days	SM	CL	LW	RIE
VOMXCUS\$	Below threshold	318	0.6041	0.6033	0.5989	0.5961
	Above threshold	188	0.7953	0.7873	0.7911	0.7845
VOMXGUS\$	Below threshold	333	0.6686	0.6719	0.6632	0.6594
	Above threshold	173	0.7091	0.6905	0.7052	0.6998

High-threshold days coincide with higher realized risk

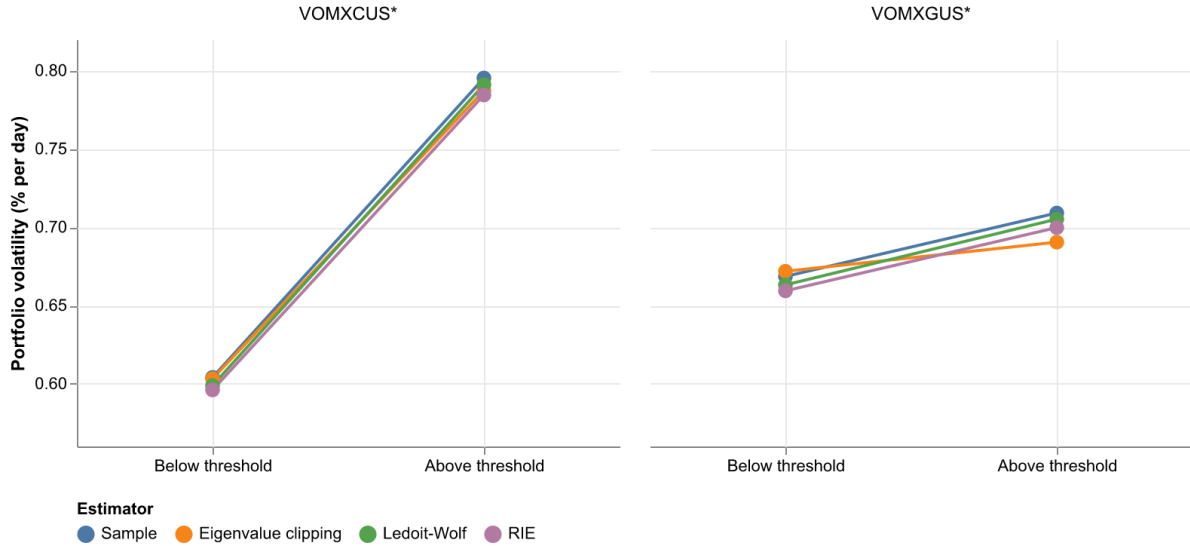


Figure 3: Holdout daily volatility conditional on the converted VOM-series thresholds. Thresholds are estimated on the training period and held fixed; no causal interpretation or regime-difference significance test is available. Proxy field definitions remain unverified.

3.3 Sensitivity to q

Recomputation from the matched 25-stock Bloomberg panel yields 100 daily returns (13 May–29 September 2026), so $q = 25/100 = 0.25$. Using the gross total-return index, the sample covariance condition number is 279.55 and the sample correlation condition number is 24.34. The largest sample-correlation eigenvalue is 4.178, above the Marchenko–Pastur upper edge 2.25; one eigenvalue lies above that edge. Eigenvalue clipping and diagonal renormalization reduce the correlation condition number to 5.33 and the covariance condition number to 128.49. Using unadjusted `PX_LAST` instead gives a correlation condition number of 24.31, a largest eigenvalue of 4.170, and a clipped correlation condition number of 5.30. Clipping reduces the correlation condition number by 78.1% and the covariance condition number by 54.0%. The largest correlation eigenvalue, 4.178, is 1.86 times the random-matrix upper noise edge, with one eigenvalue above that edge. This recent window illustrates spectral stabilization but has no subsequent holdout, so improved conditioning cannot establish lower future portfolio risk. These figures are recomputed from daily observations in the latest workbook, not from reported summary statistics.

Eigenvalue clipping improves conditioning at $q = 0.25$

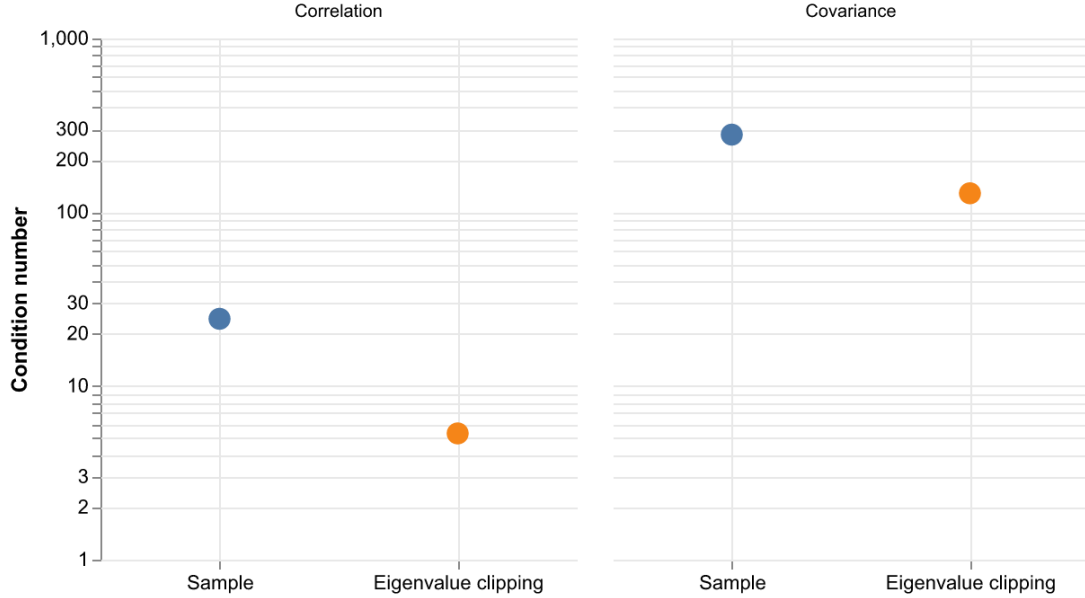


Figure 4: Sample versus clipped condition numbers for covariance and correlation matrices in the 25-stock gross-total-return panel ($q = 0.25$; logarithmic vertical scale). Better conditioning is a numerical property, not by itself an out-of-sample performance result.

At $q = 1.9$ with 38 assets and 20 training days, the sample and unmodified RIE covariance estimates are singular in all ten rolling windows and require a pseudo-inverse. Across the following 60-day test windows, mean annualized realized volatility is 16.57% for SM, 11.78% for CL, 12.12% for LW, and 144.43% for RIE (window-to-window standard deviation 320.17% for RIE). Thus the unmodified RIE implementation is unstable in this $q > 1$ regime; no extension to $q \geq 1$ is claimed. Figure 5 uses a logarithmic scale because the RIE mean is an order of magnitude larger than the other three estimates. The exceptionally large 320.17% between-window standard deviation for RIE further indicates that its mean alone understates the instability. The pseudo-inverse fallback makes these results a stress test of the full pipeline, not a comparison of four equally well-posed estimators.

The distributional sensitivity in Table 3 is visualized in Figure 6. Training marginals are rank-Gaussianized and rescaled, but evaluation remains on the unchanged holdout. RIE still has lower holdout volatility than SM under this transformation; the result does not hold against every alternative estimator, nor does it establish a ranking by the holdout-to-training risk ratio.

3.4 Fama–French factor check

On the 522 holdout dates matched to the public daily Fama–French five-factor file [8], the RIE portfolio’s estimated daily alpha is 0.000002 with $p = 0.995$. The other three estimators likewise have insignificant alphas ($p \geq 0.887$). This check does not support an abnormal-return claim. Because the factors are constructed from U.S. equities while the test portfolios are Mexican, this regression is an exploratory cross-market exposure diagnostic, not a validated Mexican-equity pricing model. Its insignificant intercepts do not establish factor neutrality.

At $q = 1.9$, this unmodified RIE implementation is unstable

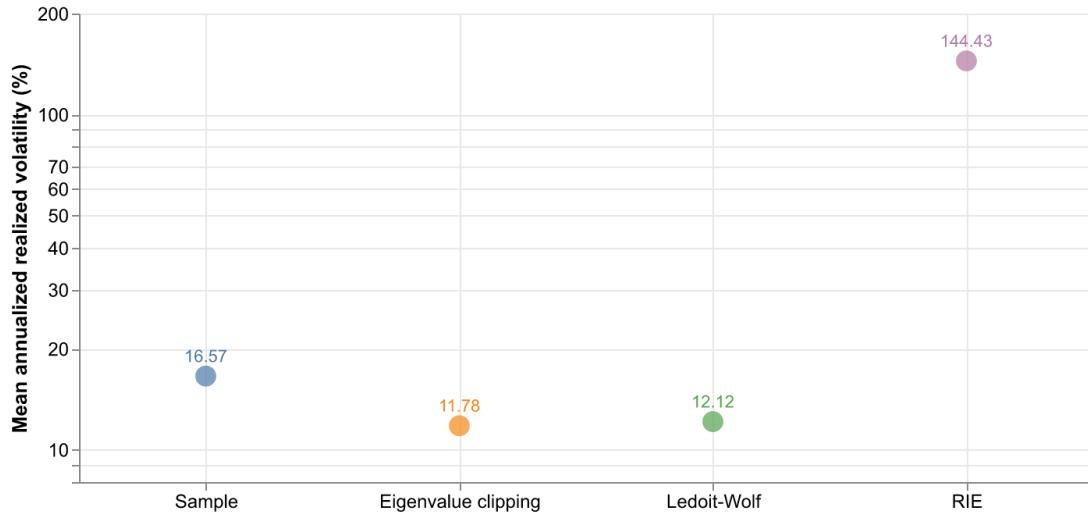


Figure 5: Mean annualized realized volatility in ten deliberately short $q = 1.9$ training windows (logarithmic vertical scale). RIE's window-to-window standard deviation is 320.17%; the other estimators' standard deviations were not preserved in the available output.

Rank-Gaussianizing training data weakens, but preserves, RIE's edge over sample

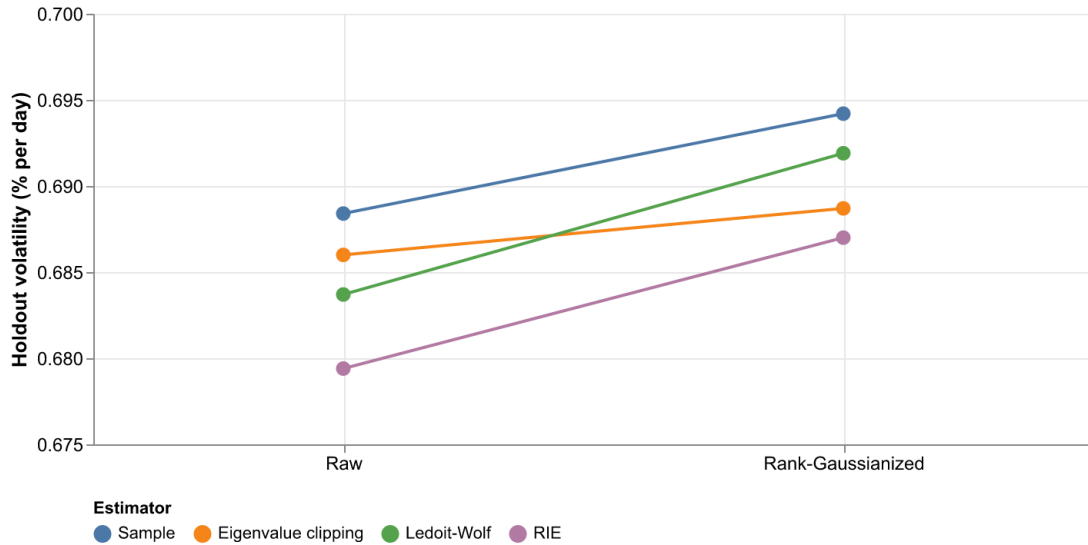


Figure 6: Holdout volatility after fitting on raw versus rank-Gaussianized training marginals. The holdout is not transformed. Truncated vertical scale highlights small differences.

4 Discussion

The primary answer is deliberately narrow. On one common BMV holdout, RIE has the lowest point-estimate realized volatility, 1.31% below SM in relative terms, but the magnitude is small and the available bootstrap tests Sharpe differences rather than volatility differences. Its Sharpe advantage over SM is not statistically distinguishable from zero. This evidence is compatible with a modest realized-risk improvement in this split; it does not establish a general out-of-sample superiority of nonlinear shrinkage.

Allocation breadth also admits no unqualified "more diversified" conclusion. RIE's inverse-Herfindahl count rises by 0.50 effective holdings (5.3%) while its Shannon breadth is unchanged at the reported precision. Moreover, the weights are unconstrained and may be short, so these indices describe weight concentration rather than literal portfolio membership. They do not measure the diversity of investors, strategies, or beliefs in the market.

The regime comparisons are consistent with higher portfolio risk on high-threshold VOM-labelled days, especially for VOMXCUS; the difference is smaller for VOMXGUS. However, ticker semantics, data fields, and currency interpretation are not documented in the export, and the exercise has no reported uncertainty test. These results motivate validation of the proxy metadata, not a claim that an identified economic regime causes a volatility increase.

The $q = 0.25$ cross-section shows that clipping can materially improve numerical conditioning, including under a gross-total-return input. A better condition number can reduce sensitivity of matrix inversion but does not demonstrate improved realized risk by itself. Conversely, the $q = 1.9$ experiment is a strong failure signal for this particular implementation: unmodified RIE retains structural singularity and its realized-risk estimates are extremely unstable. It is not a theorem about RIE in general, since the kernel estimator and pseudo-inverse fallback were not redesigned for $q > 1$.

Several limitations constrain external validity. First, the primary price source is licensed, and the universe construction is not documented in enough detail to rule out survivorship or selection bias. Second, one static train/test split does not substitute for repeated rolling-origin evaluation; no transaction costs, turnover, market impact, liquidity, short-sale frictions, or leverage limits are modeled. Third, the assumed 8% risk-free rate is not varied, and no confidence interval for the volatility difference is available. Fourth, the Bloomberg VOM definitions need independent confirmation and the 25-stock recent panel is a fixed diagnostic set, not a random sample of the 275-tab universe. Finally, the U.S. factor regression is not a country-matched pricing test. The paper's conclusion should therefore be read as evidence about a particular panel, split, implementation, and portfolio construction.

5 Conclusion

For the common 38-stock BMV holdout, nonlinear spectral cleaning is associated with a small reduction in realized minimum-variance risk and no measured reduction in either reported allocation-breadth measure relative to the sample estimator. The Sharpe-ratio difference is not statistically significant, and Shannon breadth is unchanged. The supplementary evidence shows descriptive regime dependence, numerical conditioning gains at $q = 0.25$, and a severe instability boundary for the unmodified RIE implementation at $q = 1.9$. The evidence supports considering RIE as one covariance-estimation option in the low- q setting studied, with repeated out-of-sample monitoring and explicit estimator-validity checks. It does not establish universal estimator dominance or market-wide changes in investor heterogeneity.

Data availability. The Fama–French five-factor series is available from the Kenneth French Data Library (https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html). The primary Economática prices and supplementary Bloomberg exports (the VOM/NAFTRAC/USD–MXN CSV and the 275-instrument BMV/BIVA workbook, downloaded in September 2026) are subject to data-provider terms and are not included. This working paper’s companion repository (<https://github.com/IvliCaesar/bmv-spectral-portfolio-cleaning>) contains only aggregate summary tables and charts, not daily prices, returns, security weights, or licensed exports. Those summary outputs allow readers to recreate the displayed graphics and check selected arithmetic; they do not permit re-estimation of the portfolios, bootstrap, factor regressions, regime classification, or eigenspectra. The companion repository does not include the original estimator-analysis scripts; full replication therefore requires lawful access to both licensed inputs and the complete analysis workflow.

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